



GIRRAWEE HIGH SCHOOL

Task 2

2014

MATHEMATICS

*Time allowed – 90 minutes
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL 6 questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are on laminated sheet provided.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1 , Question 2 , etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.



**GIRRAWEE HIGH SCHOOL
MATHEMATICS
2014**

FINAL MARK

HIGHER SCHOOL CERTIFICATE EXAMINATION

ANSWERS COVER SHEET

Name: _____

QUESTION	MARK	H2	H3	H4	H5	H6	H7	H8	H9
Section 1	/5								√
Part B Q6	/10				√				√
Q 7	/14				√	√			√
Q 8	/10				√	√			√
Q 9	/11				√	√			√
Q 10	/22				√	√			√
Q 11	/9				√	√		√	√
TOTAL	/81				/76	/66		/9	/81

HSC Outcomes

Mathematics

- H1 seeks to apply mathematical techniques to problems in a wide range of practical contexts.
- H2 constructs arguments to prove and justify results.
- H3 manipulates algebraic expressions involving logarithmic and exponential functions.
- H4 expresses practical problems in mathematical terms based on simple given models.
- H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems.
- H6 uses the derivative to determine the features of the graph of a function.
- H7 uses the features of a graph to deduce information about the derivative.
- H8 uses techniques of integration to calculate areas and volumes.
- H9 communicates using mathematical language, notation, diagrams and graphs.

**GIRRAWEE HIGH SCHOOL
MATHEMATICS**

YEAR 12 HSC

Task 2 2014

Time Allowed: 90 minutes

INSTRUCTIONS TO STUDENTS

- Attempt **ALL 6** questions.
- Circle the best response for the questions in **Part A**
- Start each question in **Part B** on a new page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.

PART A (5 marks)

For questions 1-5 circle the best response from the following:

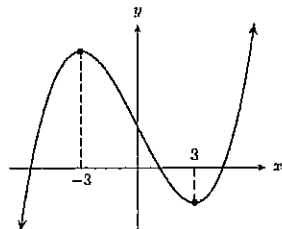
Question 1. What are the x -coordinates of the two stationary points to the curve $y = 5 + 3x^3 - 2x^4$?

- (A) $x = 0, x = \frac{2}{3}$
- (B) $x = 0, x = \frac{3}{2}$
- (C) $x = 0, x = \frac{8}{9}$
- (D) $x = 0, x = \frac{9}{8}$

Question 2. For which values of x is the curve $f(x) = -x^2 + 4x + 5$ increasing?

- (A) $x > 2$ (B) $x < 2$ (C) $x \geq -2$ (D) $x \leq -2$

Question 3. From the graph of $y = f(x)$, when is $f'(x)$ negative?

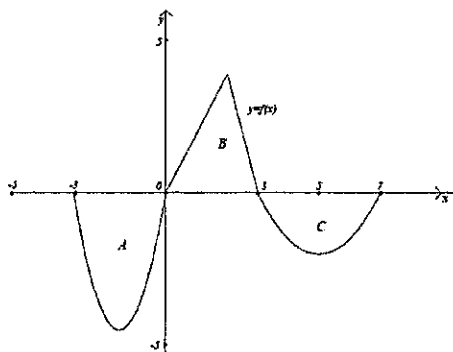


- (A) $x < -3$ or $x > 3$ (B) $-3 < x < 3$ (C) $x \leq -3$ or $x \geq 3$ (D) $-3 \leq x \leq 3$

Question 4. What is the value of $\int_3^5 (x+1)^{-2} dx$?

- (A) $\frac{1}{12}$ (B) $\frac{1}{24}$ (C) $\frac{5}{24}$ (D) $-\frac{5}{24}$

Question 5.



The graph on the right shows the curve $y = f(x)$. The shaded areas are bounded by $y = f(x)$ and the x axis. Shaded area A is 9 square units, shaded area B is 6 square units and shaded area C is 5 square units.

The value of $\int_{-3}^7 f(x) dx$

- (A) 8 (B) -8 (C) 20 (D) -20

PART B

QUESTION 6. (10 marks)

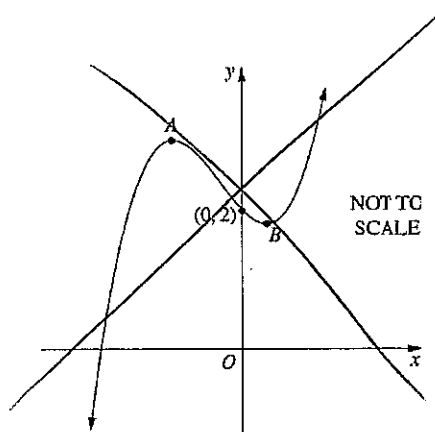
- (a) Show that the curve $y = 2x - \frac{3}{x-1}$ is increasing for all values of x . 2
- (b) The curve $y = x^3 + ax^2 + bx + 5$ has a stationary point at $(2, -3)$. Find the values of a and b . 3
- (c) If $y = \frac{x}{x-1}$, show that $\frac{2yy'}{x} + y'' = 0$. 3
- (d) For what values of x is the curve $y = 2x^3 + 3x^2 - 12x + 8$ concave up? 2

QUESTION 7 (14 marks)

- (a) Show that the curve $y = x^4 - x + 1$ has no points of inflexion. 3
- (b) The function $y = x^3 - 3x^2 - 9x + 1$ is defined in the domain $-4 \leq x \leq 5$.
- (i) Find the coordinates of any turning points and determine their nature. 4
- (ii) Find the coordinates of any points of inflexion. 2
- (iii) Draw a neat sketch of the curve. 3
- (iv) Determine the minimum value of the function y , in the domain $-4 \leq x \leq 5$ 2

Question 8. (10 marks)

(a)

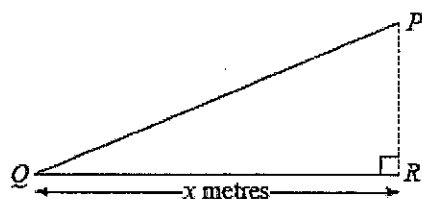


The diagram shows a sketch of the curve $y = 6x^2 - x^3$. The curve cuts the x -axis at L , and has a local maximum at M and a point of inflection at N .

- (i) Find the coordinates of L . 1
- (ii) Find the coordinates of M . 1
- (iii) Find the coordinates of N . 1

Question 8 continued

(b)



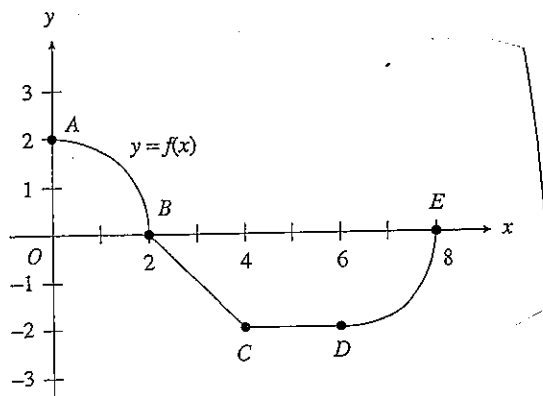
A wire of length 5 metres is to be bent to form the hypotenuse and base of a right-angled triangle PQR, as shown in the diagram. Let the length of the base QR be x metres.

- (i) What is the length of the hypotenuse PQ in terms of x ? 1
- (ii) Show that the area of the triangle PQR is $\frac{1}{2}x\sqrt{25-10x}$ square metres. 2
- (iii) What is the maximum possible area of the triangle? 4

QUESTION 9. (11marks)

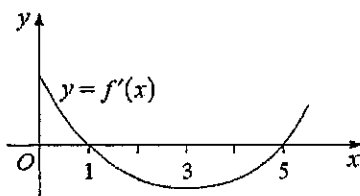
- (a) Find the primitive function of $\frac{2}{\sqrt{x}} + \frac{1}{\sqrt[3]{x^2}}$. 3
- (b) Find $f(x)$, given that $f'(x) = 6x^2 - 6x + 5$ and that the curve passes through the point (2,13). 3

(c)



The graph of the function f consists of a quarter circle AB, a straight line segment BC, a horizontal straight line segment CD, and a quarter circle DE as shown above. Evaluate $\int_0^8 f(x)dx$. 3

(d)



The diagram shows the graph of the gradient function of the curve $y = f(x)$. For what values of x does $f(x)$ have a local minimum? Justify your answer. 2

Question 10.(22 marks)

(a) Find each indefinite integral:

(i) $\int (3x^2 - 2) dx$ **1**

(ii) $\int (\sqrt{x} - 2)(\sqrt{x} + 2) dx$ **3**

(iii) $\int \frac{dx}{(2x+3)^2}$ **3**

(iv) $\int \frac{\sqrt{x}-1}{2\sqrt{x}} dx$ **3**

(b) Evaluate:

(i) $\int_0^1 (3x^6 + 1) dx$ **2**

(ii) $\int_0^3 \frac{dx}{(2x-3)^2}$ **3**

(iii) $\int_2^7 \frac{dx}{\sqrt{x+2}}$ **3**

(iv) (i) Find $\frac{d}{dx}(x^2 + 1)^5$. **2**

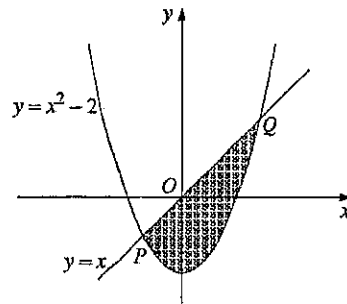
(ii) Hence find $\int 10x(x^2 + 1)^4 dx$ **2**

Question 11.(9 marks)

- (a) Calculate the area enclosed by the curve $y = 4x^2$, the y axis and the lines $y = 1$, $y = -4$, lying in the first quadrant.

4

- (b) The diagram shows the graphs of $y = x^2 - 2$ and $y = x$.



- (i) Find the x values of the points of intersection, P and Q.
- (ii) Calculate the area of the shaded region.

2

3

End of Examination

Year 12 mathematics Task 2 2014 solutions.

Part A

- ① D ③ B ⑤ B
② B ④ A

Part B

Question 6

a) $y = 2x - \frac{3}{x-1}, x \neq 1$

$$y' = 2 + 3(x-1)^{-2}$$

$$= 2 + \frac{3}{(x-1)^2}$$

$(x-1)^2 > 0$ for all values of x .

$\therefore y' > 0$ for all values of x .

$\therefore$ The curve is increasing for all values of x

(2)

b) $y = x^3 + ax^2 + bx + 5$

$$y' = 3x^2 + 2ax + b$$

Since $(2, -3)$ is a stationary point, when $x=2, y'=0$

$$\therefore 0 = 12 + 4a + b$$

$$\therefore 4a + b = -12 \quad \text{--- (1)}$$

Since the curve passes through $(2, -3)$,

in $y = x^3 + ax^2 + bx + 5$,

$$\left. \begin{matrix} x=2 \\ y=-3 \end{matrix} \right\} \quad -3 = 8 + 4a + 2b + 5$$

$$\therefore 4a + 2b = -16 \quad \text{--- (2)}$$

$$4a + b = -12 \quad \text{--- (1)}$$

$$4a + 2b = -16 \quad \text{--- (2)}$$

$$\text{①} - \text{②} \quad -b = 4 \quad \therefore b = -4$$

$$\therefore a = -2 \quad \text{--- (3)}$$

c) $y = \frac{x}{x-1}$

$$y' = \frac{1(x-1) - x}{(x-1)^2} = \frac{-1}{(x-1)^2}$$

$$y'' = 2(x-1)^{-3} = \frac{2}{(x-1)^3}$$

LHS $\frac{2yy'}{x} + y'' =$

$$= \frac{2}{x} \times \frac{x}{x-1} \times \frac{-1}{(x-1)^2} + \frac{2}{(x-1)^3}$$

$$= 0 = \text{RHS} \quad \text{--- (3)}$$

d) $y = 2x^3 + 3x^2 - 12x + 8$

The curve is concave up $\Rightarrow y'' > 0$

$$y' = 6x^2 + 6x - 12$$

$$y'' = 12x + 6 > 0 \Rightarrow x > -\frac{1}{2}$$

(2)

Question 7

a) $y = x^4 - x + 1$

$$y' = 4x^3 - 1$$

$$y'' = 12x^2$$

At points of inflexion, $y'' = 0$

$$\therefore 12x^2 = 0 \quad \therefore x = 0$$

when $x = 0$, $y = 1$

$\therefore (0, 1)$ is a possible point of inflexion.

Concavity test:

x	-0.1	0	0.1
y''	+	0	+

There is no change of concavity.

$\therefore (0, 1)$ is not a p.o. inflexion.

$\therefore$ The curve has no p.o. inflexion.

(3)

b) $y = x^3 - 3x^2 - 9x + 1$

$$y' = 3x^2 - 6x - 9$$

At stationary points $y' = 0$

$$\therefore 3(x^2 - 2x - 3) = 0$$

$$3(x - 3)(x + 1) = 0$$

$\therefore$ stationary points at $x = -1, 3$

When $x = -1$, $y = (-1)^3 - 3(-1)^2 - 9(-1) + 1$
 $= 6$

$x = 3$, $y = -26$

$$y'' = 6x - 6$$

at $x = -1$, $y'' < 0 \therefore$ maximum

at $x = 3$, $y'' > 0 \therefore$ minimum

$\therefore$ maximum at $(-1, 6)$

minimum at $(3, -26)$ (4)

possible

(i) For points of inflexion

$$y'' = 0 \Rightarrow 6(x - 1) = 0$$

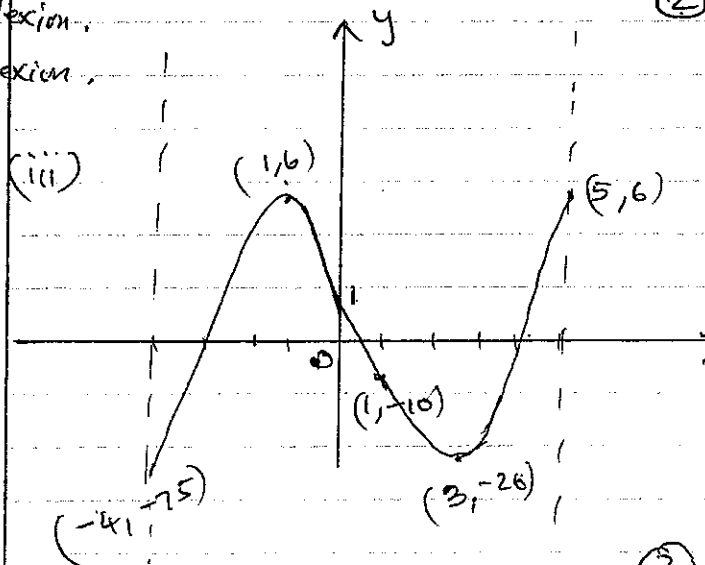
$$\therefore x = 1$$

Concavity test:

x	-0	1	2
y''	-	0	+

$\therefore$ concavity change occurs when $x = 1$, $y = -10$

$\therefore (1, -10)$ is a point of inflexion (2)



(iii)

(2)

(iv) minimum value = -26

(2)

Question 8

a) (i) $y = 6x^2 - x^3$
 $= x^2(6 - x)$

At L, $y = 0$

$\therefore x = 0 \text{ or } 6$

$\therefore$ coordinates of L are (6, 0) ①

(ii) $\frac{dy}{dx} = 12x - 3x^2$
 $= 3x(4 - x)$

at M, $\frac{dy}{dx} = 0$

$\therefore 3x(4 - x) = 0$

$\therefore x = 0 \text{ or } 4$

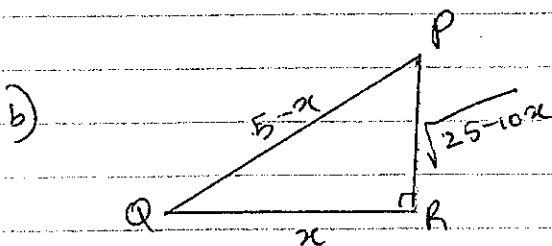
When $x = 4$, $y = 6 \cdot 4^2 - 4^3$
 $= 32$

$\therefore N = (4, 32)$ ①

(iii) $\frac{d^2y}{dx^2} = 12 - 6x = 6(2 - x)$

At N, $\frac{d^2y}{dx^2} = 0 \Rightarrow x = 2$

$\therefore N = (2, 16)$ ①



(i) $PQ = (5 - x)m$

(ii) $PR^2 = (5 - x)^2 - x^2$
 $= 25 - 10x$

$\therefore PR = \sqrt{25 - 10x}, (PR > 0)$ ①

$\therefore$ Area of ΔPQR

$= \frac{1}{2} \times b \times h$

$= \frac{1}{2} \times x \times \sqrt{25 - 10x}$

$= \frac{1}{2} x \sqrt{25 - 10x} \text{ m}^2$ ②

(iii) $A = \frac{x}{2} \sqrt{25 - 10x}$

$u = \frac{1}{2}x \quad v = (25 - 10x)^{\frac{1}{2}}$

$u' = \frac{1}{2} \quad v' = \frac{1}{2}(25 - 10x)^{-\frac{1}{2}} \times -10$

$v' = -5(25 - 10x)^{-\frac{1}{2}}$

$v' = \frac{-5}{\sqrt{25 - 10x}}$

$y' = u'v + uv'$

$= \frac{\sqrt{25 - 10x}}{2} - \frac{5x}{2\sqrt{25 - 10x}}$

$= \frac{25 - 10x - 5x}{2\sqrt{25 - 10x}}$

$= \frac{25 - 15x}{2\sqrt{25 - 10x}}$

At stationary points, $\frac{dA}{dx} = 0$

$\therefore \frac{-15x + 25}{2\sqrt{25 - 10x}} = 0$

$\therefore 15x = 25 \Rightarrow x = \frac{5}{3}$

Concavity test

x	1	$\frac{5}{3}$	2
$\frac{dA}{dx}$	+	0	-

$\therefore$ Concavity changes ④

$\therefore$ A maximum occurs at

$x = \frac{5}{3}$

$\therefore A_{\max} = \frac{1}{2} \times \frac{5}{3} \times \sqrt{25 - 10 \times \frac{5}{3}} = \frac{25\sqrt{5}}{18} \text{ m}^2$

Question 9

$$\begin{aligned}
 a) \quad f'(x) &= \frac{2}{\sqrt{x}} + \frac{1}{\sqrt[3]{x^2}} \\
 &= 2x^{-\frac{1}{2}} + x^{-\frac{2}{3}} \\
 \therefore f(x) &= \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + \frac{x^{\frac{1}{3}}}{\frac{1}{3}} + C \\
 &= 4x^{\frac{1}{2}} + 3x^{\frac{1}{3}} + C \\
 &= 4\sqrt{x} + 3\sqrt[3]{x} + C
 \end{aligned}$$

Question 10

d) At $x=5$, $f'(x)=0$
 when $x < 5$, $f'(x) < 0$
 when $x > 5$, $f'(x) > 0$
 $\therefore$ A local minimum at $x=5$

(2)

$$\begin{aligned}
 b) \quad f'(x) &= 6x^2 - 6x + 5 \\
 f(x) &= 6 \cdot \frac{x^3}{3} - 6 \cdot \frac{x^2}{2} + 5x + C \\
 &= 2x^3 - 3x^2 + 5x + C
 \end{aligned}$$

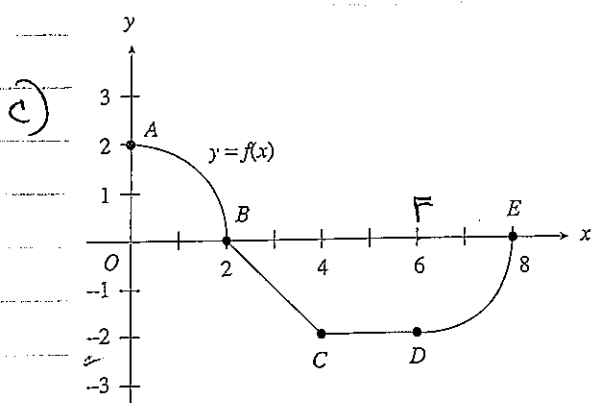
Sub in $f(2) = 13 \Rightarrow$

$$13 = 16 - 12 + 10 + C$$

$$\therefore C = -1$$

$$\therefore f(x) = 2x^3 - 3x^2 + 5x - 1$$

(3)



Let $F = (6, 0)$

$$\begin{aligned}
 \therefore \int_0^8 f(x) dx &= (\text{area } \frac{1}{4} \text{ circle } AB) \\
 &\quad - \text{Area of } \frac{1}{4} \text{ circle } DE \\
 &\quad - \text{area of trapezium } BCDF \\
 &= 0 - \frac{1}{2}(2+4) \cdot 2 = -6
 \end{aligned}$$

(3)

$$a) (i) \int (3x^2 - 2) dx = x^3 - 2x + C$$

(1)

$$\begin{aligned}
 (ii) \quad \int (\sqrt{x+2})(\sqrt{x+2}) dx \\
 = \int (x-4) dx = \frac{x^2}{2} - 4x + C
 \end{aligned}$$

(3)

$$\begin{aligned}
 (iii) \quad \int \frac{dx}{(2x+3)^2} &= \int (2x+3)^{-2} dx \\
 &= \frac{(2x+3)^{-1}}{-1 \times 2} + C \\
 &= -\frac{1}{2(2x+3)} + C
 \end{aligned}$$

(3)

$$\begin{aligned}
 (iv) \quad \int \frac{\sqrt{x}-1}{2\sqrt{x}} dx \\
 = \int \frac{1}{2} - \frac{1}{2\sqrt{x}} dx \\
 = \frac{x}{2} - \frac{1}{2} \cdot \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C \\
 = \frac{x}{2} - \sqrt{x} + C
 \end{aligned}$$

Question 10

$$\text{a) (i) } \int (3x^2 - 2) dx = x^3 - 2x + C \quad (1)$$

$$\begin{aligned} \text{(ii) } \int (\sqrt{x} - 2)(\sqrt{x} + 2) dx \\ = \int (x - 4) dx = \frac{x^2}{2} - 4x + C \end{aligned} \quad (2)$$

$$\begin{aligned} \text{(iii) } \int \frac{dx}{(2x+3)^2} &= \int (2x+3)^{-2} dx \\ &= -\frac{1}{2(2x+3)} + C \end{aligned}$$

$$\begin{aligned} \text{(iv) } \int \frac{\sqrt{x} - 1}{2\sqrt{x}} dx &= \int \left(\frac{1}{2} - \frac{1}{2\sqrt{x}} \right) dx \\ &= \frac{x}{2} - \frac{2\sqrt{x}^{\frac{1}{2}}}{\frac{1}{2}} + C \\ &= \frac{x}{2} - \sqrt{x} + C \end{aligned} \quad (3)$$

$$\begin{aligned} \text{b) (i) } \int_0^1 (3x^6 + 1) dx &= \left[\frac{3x^7}{7} + x \right]_0^1 \\ &= 1\frac{3}{7} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{(ii) } \int_0^3 \frac{dx}{(2x-3)^2} &= \left[\frac{(2x-3)^{-1}}{-2} \right]_0^3 \\ &= -\frac{1}{2} \left[\frac{1}{2x-3} \right]_0^3 = -\frac{1}{2} \left[\frac{1}{3} + \frac{1}{3} \right] \\ &= -\frac{1}{3} \end{aligned} \quad (3)$$

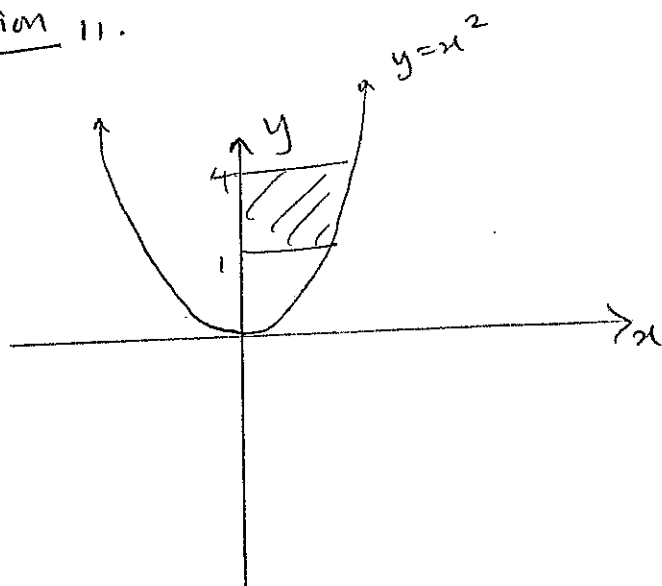
$$\begin{aligned} \text{(ii) } \int_2^3 \frac{dx}{\sqrt{x+2}} &= \int_2^3 (x+2)^{-\frac{1}{2}} dx \\ &= \left[\frac{(x+2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_2^3 = 2 \left[\sqrt{x+2} \right]_2^3 \\ &= 2 [3 - 2] = 2 \end{aligned} \quad (3)$$

$$\begin{aligned} \text{(iv) (a) } \frac{d}{dx} (x^2 + 1)^5 &= 5(x^2 + 1)^4 \times 2x \\ &= 10x(x^2 + 1)^4 \end{aligned}$$

$$\text{(b) } \therefore \int 10x(x^2 + 1)^4 dx = (x^2 + 1)^5 + C$$

Question 11.

a)



$$y = 4x^2 \quad \therefore x = \frac{\sqrt{y}}{2}$$

$$A = \int_1^4 x dy = \int_1^4 \frac{\sqrt{y}}{2} dy$$

$$= \left[\frac{y^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^4 = \frac{2}{3} y^{\frac{3}{2}} \Big|_1^4$$

(4)

b) (i) At points of intersection,

$$x^2 - 2 = x$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$\therefore x = -1, 2$$

(2)

(ii) Area of shaded region,

$$= \int_{-1}^2 x - (x^2 - 2) dx$$

$$= \int_{-1}^2 (-x^2 + x + 2) dx$$

$$= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2$$

$$= \left[-\frac{8}{3} + 2 + 4 \right] - \left[\frac{1}{3} - \frac{1}{2} - 2 \right] = 4\frac{1}{6}$$

Question 10

$$a) (i) \int (3x^2 - 2) dx = x^3 - 2x + C \quad (1)$$

$$(ii) \int (\sqrt{x} - 2)(\sqrt{x} + 2) dx \\ = \int (x - 4) dx = \frac{x^2}{2} - 4x + C \quad (2)$$

$$(iii) \int \frac{dx}{(2x+3)^2} = \int (2x+3)^{-2} dx \\ = -\frac{1}{2(2x+3)} + C$$

$$(iv) \int \frac{\sqrt{x} - 1}{2\sqrt{x}} = \int \left(\frac{1}{2} - \frac{1}{2\sqrt{x}} \right) dx \\ = \frac{x}{2} - \frac{2\sqrt{x}^{\frac{1}{2}}}{\frac{1}{2}} + C \\ = \frac{x}{2} - \sqrt{x} + C \quad (3)$$

$$b) (i) \int_0^1 (3x^6 + 1) dx = \left[\frac{3x^7}{7} + x \right]_0^1 \\ = 1\frac{3}{7} \quad (2)$$

$$(ii) \int_0^3 \frac{dx}{(2x-3)^2} = \left[\frac{(2x-3)^{-1}}{-2} \right]_0^3 \\ = -\frac{1}{2} \left[\frac{1}{2x-3} \right]_0^3 = -\frac{1}{2} \left[\frac{1}{3} + \frac{1}{3} \right] \\ = -\frac{1}{3} \quad (3)$$

$$(ii) \int_2^3 \frac{dx}{\sqrt{x+2}} = \int_2^3 (x+2)^{-\frac{1}{2}} dx \\ = \left[\frac{(x+2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_2^3 = 2 \left[\sqrt{x+2} \right]_2^3 \\ = 2 [3 - 2] = 2 \quad (3)$$

$$(iv) (a) \frac{d}{dx} (x^2 + 1)^5 = 5(x^2 + 1)^4 \times 2x \\ = 10x(x^2 + 1)^4$$

$$(b) \therefore \int 10x(x^2 + 1)^4 dx = (x^2 + 1)^5 + C$$